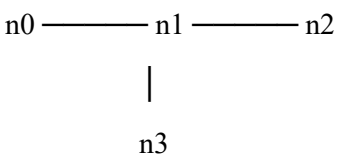
Simulation of Simple Network using NS3

Deadline: 24/8/26, 26/8/26

Create and simulate a simple computer network with multiple nodes, point-to-point links, and traffic flows using **NS-3**, and measure basic network performance using the below requirements.

Requirements:
Create a network consisting of four nodes (n0, n1, n2, and n3) connected using three point-to-point links as shown below.



Configure the links with a data rate of 5 Mbps and a propagation delay of 2 ms. Assign appropriate IP addresses to all network interfaces and enable IP routing.

Generate the following two UDP traffic flows:

- Flow 1: n0 → n2 at 1 Mbps
- Flow 2: n3 → n0 at 500 Kbps

Run the simulation for 10 seconds.

Do the following.

1. Create the four nodes and three point-to-point links.
2. Configure the data rate and propagation delay of the links.
3. Install the Internet protocol stack on all nodes.
4. Assign IP addresses to the network interfaces.
5. Configure routing between the nodes.
6. Create the two UDP traffic flows.
7. Run the simulation for 10 seconds.
8. Calculate and display:
    - Total bytes received
    - Throughput of each flow
    - Average packet latency without using FlowMonitor.

9. Compare the performance of the two traffic flows.

```
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"

using namespace ns3;

NS_LOG_COMPONENT_DEFINE("SimpleNetwork");

int main(int argc, char *argv[])
{
   // Create 4 nodes
   NodeContainer nodes;
   nodes.Create(4);

   // Create links
   PointToPointHelper p2p;
   p2p.SetDeviceAttribute("DataRate", StringValue("5Mbps"));
   p2p.SetChannelAttribute("Delay", StringValue("2ms"));

   // n0-n1, n1-n2, n3-n2
   NetDeviceContainer d01 = p2p.Install(nodes.Get(0), nodes.Get(1));
   NetDeviceContainer d12 = p2p.Install(nodes.Get(1), nodes.Get(2));
   NetDeviceContainer d32 = p2p.Install(nodes.Get(3), nodes.Get(2));

   // Install Internet stack
```

```
InternetStackHelper internet;
internet.Install(nodes);

// Assign IP addresses
Ipv4AddressHelper address;

address.SetBase("10.1.1.0", "255.255.255.0");
Ipv4InterfaceContainer i01 = address.Assign(d01);

address.SetBase("10.1.2.0", "255.255.255.0");
Ipv4InterfaceContainer i12 = address.Assign(d12);

address.SetBase("10.1.3.0", "255.255.255.0");
Ipv4InterfaceContainer i32 = address.Assign(d32);

// Enable routing
Ipv4GlobalRoutingHelper::PopulateRoutingTables();

// -------------------------
// Flow 1: n0 -> n2
// -------------------------

uint16_t port1 = 5000;

PacketSinkHelper sink1(
    "ns3::UdpSocketFactory",
    InetSocketAddress(Ipv4Address::GetAny(), port1));

ApplicationContainer sinkApp1 =
```

```
    sink1.Install(nodes.Get(2));

sinkApp1.Start(Seconds(1.0));
sinkApp1.Stop(Seconds(10.0));

OnOffHelper source1(
    "ns3::UdpSocketFactory",
    InetSocketAddress(i12.GetAddress(1), port1));

source1.SetAttribute(
    "DataRate",
    StringValue("1Mbps"));

source1.SetAttribute(
    "PacketSize",
    UintegerValue(512));

ApplicationContainer sourceApp1 =
    source1.Install(nodes.Get(0));

sourceApp1.Start(Seconds(2.0));
sourceApp1.Stop(Seconds(10.0));

// ------------------------
// Flow 2: n3 -> n0
// ------------------------

uint16_t port2 = 6000;
```

```
PacketSinkHelper sink2(
    "ns3::UdpSocketFactory",
    InetSocketAddress(Ipv4Address::GetAny(), port2));

ApplicationContainer sinkApp2 =
    sink2.Install(nodes.Get(0));

sinkApp2.Start(Seconds(1.0));
sinkApp2.Stop(Seconds(10.0));

OnOffHelper source2(
    "ns3::UdpSocketFactory",
    InetSocketAddress(i01.GetAddress(0), port2));

source2.SetAttribute(
    "DataRate",
    StringValue("500Kbps"));

source2.SetAttribute(
    "PacketSize",
    UintegerValue(512));

ApplicationContainer sourceApp2 =
    source2.Install(nodes.Get(3));

sourceApp2.Start(Seconds(3.0));
sourceApp2.Stop(Seconds(10.0));

// Run simulation
```

```
    Simulator::Stop(Seconds(10.0));
    Simulator::Run();

    // Display received packets
    Ptr<PacketSink> sinkPtr1 =
        DynamicCast<PacketSink>(sinkApp1.Get(0));

    Ptr<PacketSink> sinkPtr2 =
        DynamicCast<PacketSink>(sinkApp2.Get(0));

    std::cout << "Flow 1 (n0 -> n2): "
              << sinkPtr1->GetTotalRx()
              << " bytes received" << std::endl;

    std::cout << "Flow 2 (n3 -> n0): "
              << sinkPtr2->GetTotalRx()
              << " bytes received" << std::endl;

    Simulator::Destroy();

    return 0;
}
```

Explanation of each Step

1. Include NS-3 libraries

```
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
```

#include "ns3/point-to-point-module.h"

#include "ns3/applications-module.h"

These provide the functions needed for:

- Simulation time
- Nodes
- Internet/IP
- Links
- Applications/traffic

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2. Create nodes

NodeContainer nodes;

nodes.Create(4);

Creates **4 nodes**:

n0 n1 n2 n3

A node can represent a **computer, router, or server**.

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3. Create links

PointToPointHelper p2p;

p2p.SetDeviceAttribute("DataRate", StringValue("5Mbps"));

p2p.SetChannelAttribute("Delay", StringValue("2ms"));

This says:

- Link speed = **5 Mbps**
- Link delay = **2 ms**

Then:

p2p.Install(nodes.Get(0), nodes.Get(1));

p2p.Install(nodes.Get(1), nodes.Get(2));

p2p.Install(nodes.Get(3), nodes.Get(2));

Creates:

n0 ——— n1 ——— n2

|

n3

---

4. Install Internet protocol

InternetStackHelper internet;

internet.Install(nodes);

This installs the Internet protocol stack, such as:

IP

UDP/TCP

Routing

on all four nodes.

---

5. Give IP addresses

For example:

address.SetBase("10.1.1.0", "255.255.255.0");

Ipv4InterfaceContainer i01 = address.Assign(d01);

The link between **n0 and n1** gets an IP network:

10.1.1.0/24

Similarly:

n0 ——— n1

10.1.1.1 10.1.1.2

n1 ——— n2

10.1.2.1 10.1.2.2

n3 ——— n2

10.1.3.1 10.1.3.2

So now the nodes can communicate using IP addresses.

---

6. Create traffic flows

This is the important part.

Flow 1: n0 → n2

PacketSinkHelper sink1(...);

sink1.Install(nodes.Get(2));

n2 is made the **receiver**.

Then:

OnOffHelper source1(...);

source1.Install(nodes.Get(0));

n0 becomes the **sender**.

So:

n0 ——> n1 ——> n2

Sender Receiver

The source sends:

Data rate = 1 Mbps

Packet size = 512 bytes

Flow 2: n3 → n0

Similarly:

n3 ——> n2 ——> n1 ——> n0

Sender Receiver

Data rate:

500 Kbps

Therefore, the simulation has **two simultaneous traffic flows**.

---

7. Run the simulation

Simulator::Stop(Seconds(10.0));

Simulator::Run();

NS-3 runs the network for **10 seconds**.

During this time:

n0 → n2

n3 → n0

packets are transmitted through the links.

8. Measure received data

sinkPtr1->GetTotalRx()

This tells us **how many bytes n2 received**.

Similarly:

sinkPtr2->GetTotalRx()

tells us **how many bytes n0 received**.

Example output:

Flow 1 (n0 -> n2): 950000 bytes received

Flow 2 (n3 -> n0): 470000 bytes received